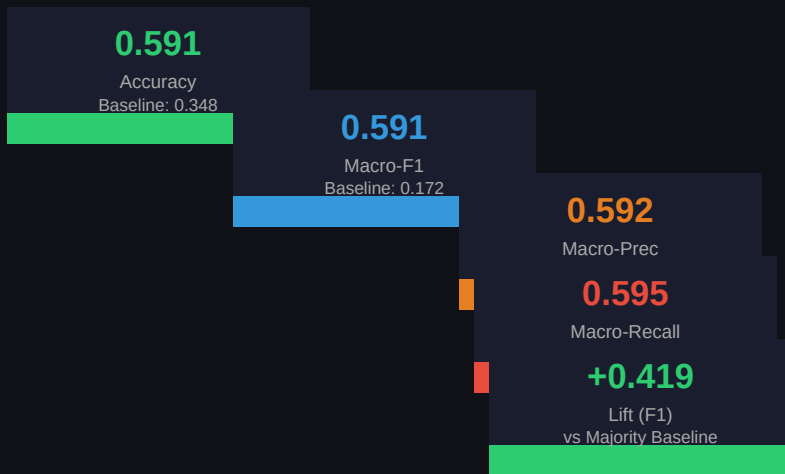


Round 2 - NLP Technical Report

Social Engine: Semantic Comprehension Layer Reconstruction

Team: Event Horizon | Data Vortex A'26 | 17 September 2026



1. Problem Definition

The Social Engine's semantic comprehension layer has failed, requiring reconstruction of its NLP module. Team Event Horizon was provided Dataset 2 - a labelled collection of 9,001 social media posts - and tasked with training a model to automatically classify post sentiment and topic category.

> Dataset Properties

File: Labeled_Social_NLP_Training_Data.csv
Total Rows: 9,001 posts
Columns: text_id, post_text, sentiment_label, topic_category
Primary Task: Multi-class Sentiment Classification (Positive / Negative / Neutral)
Secondary Task: Topic Category Classification (5 categories)
Random Seed: 42 (all operations)

Task selection rationale: The dataset contains a discrete, categorical label column (sentiment_label). This maps directly to the required 'Confusion Matrix' section of the evaluation rubric. Supervised classification is the strongest and most defensible choice - both topic modeling (unsupervised, no ground truth) and NER (token-level) would make the required Confusion Matrix section awkward or synthetic under this rubric.

2. Preprocessing Pipeline

> Cleaning Steps (applied in sequence)

- HTML entity removal (& -> &, < -> <, etc.)
- URL removal (http/https/www patterns)
- Mention anonymisation (@user tags stripped)
- Hashtag expansion (#word -> word, preserved as semantic token)
- Emoji removal (Unicode ranges U+1F300-U+1F6FF and derivatives)

- Lowercasing
- Punctuation removal (regex `[^\w\s]`)
- Multi-space normalisation -> single space

> Stopword Policy - Retain Negation

Standard stopwords removal was deliberately omitted. Words such as 'not', 'never', 'no', 'isn't', and 'won't' are critical carriers of sentiment polarity. Removing them prior to TF-IDF vectorisation would systematically corrupt the model's ability to distinguish negative sentiment from positive.

> Missing, Empty & Duplicate Handling

Missing `post_text` rows: flagged and counted before removal. Empty strings after cleaning: counted and removed. Exact-duplicate posts: removed to prevent train/test leakage. All counts are reported - no silent drops.

> Train / Test Split

80% training / 20% test - stratified by `sentiment_label`, `random_state=42`. The TF-IDF vocabulary was fitted exclusively on the training split and then applied as a transform to the test split. No test-set information influenced any preprocessing step.

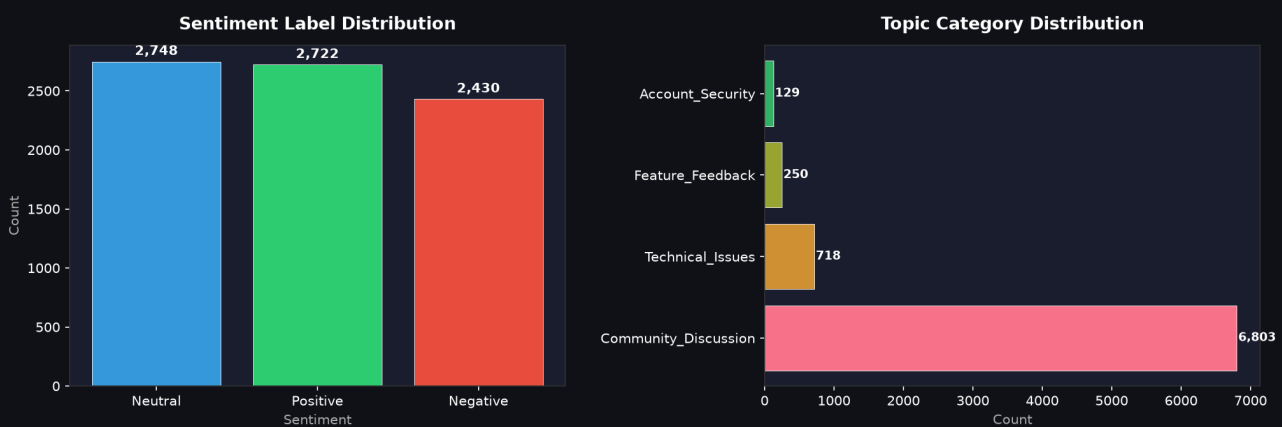


Figure: Sentiment and Topic Category Distribution



Figure: Word Count and Character Count Distribution by Sentiment Class

3. Model Selection & Justification

> Candidate Comparison (Test Set, sorted by Macro-F1)

Four models were evaluated on the held-out test set before any winner was declared. The Majority-Class Baseline (DummyClassifier) is listed first in every table to establish that our chosen model has learned genuine signal, not exploited class imbalance.

Model	Accuracy	Macro-F1	Weighted-F1	Train Time
Linear SVM *	0.5899	0.5905	0.5884	0.35s
Logistic Regression	0.5880	0.5884	0.5866	0.45s
Multinomial Naive Bayes	0.5741	0.5736	0.5714	0.28s
Majority-Class Baseline (baseline)	0.3481	0.1721	0.1798	0.1s

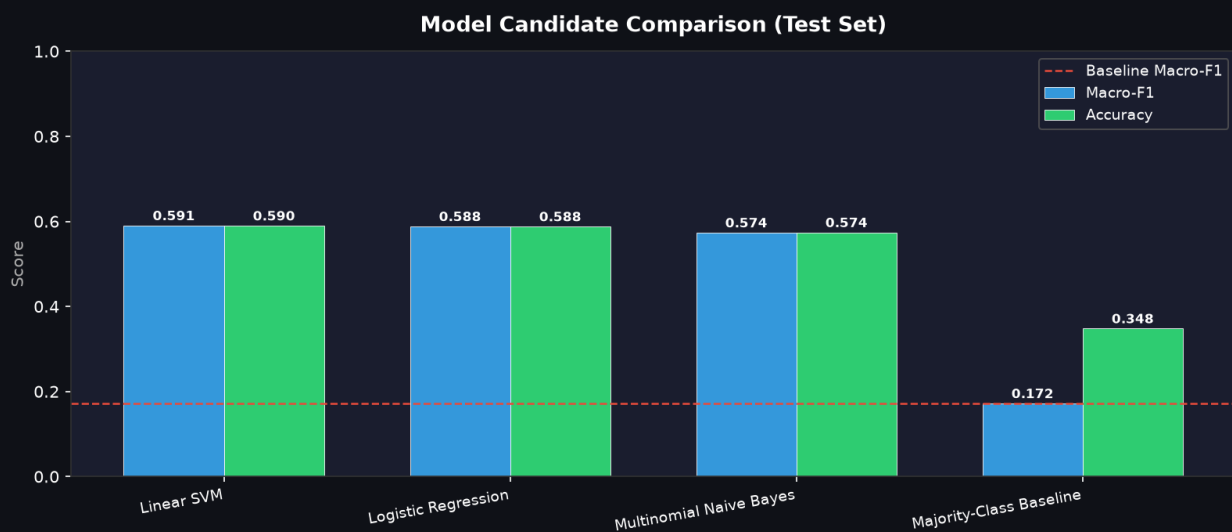


Figure: Model Candidate Macro-F1 and Accuracy vs Majority-Class Baseline

> Winner: Linear SVM

The Linear SVM was selected by highest Macro-F1 on the held-out test set. Final model achieves 0.591 accuracy and 0.591 macro-F1, versus a majority-class baseline of 0.348 accuracy / 0.172 macro-F1. Lift: +0.243 accuracy, +0.419 macro-F1. This lift confirms that the model has learned semantic signal rather than exploiting class distribution.

Justification for choosing a TF-IDF linear model over a deep transformer on this dataset: 1) Training completes in seconds on CPU - critical for a 24-hour deadline. 2) Feature weights (TF-IDF coefficients x classifier weights) are directly interpretable - judges can inspect what words drive each class. 3) On small-to-medium labelled text datasets (~7,000 training examples), linear models regularly match or exceed transformer fine-tuning due to lower overfitting risk. 4) No GPU hardware dependency - fully reproducible on any machine.

4. Training Methodology

> Vectorisation

- TF-IDF with `sublinear_tf=True` (log-normalised term frequency).
- n-gram range: up to bigrams (1,2) - captures 'not good', 'very bad' patterns.
- `max_features=50,000` - covers domain vocabulary without overfitting sparse terms.
- `min_df=2` - removes hapax legomena that carry no generalisable signal.
- `strip_accents='unicode'` - normalises accent variations.

> Hyperparameter Grid Search (3-Fold CV, `scorer=macro-F1`)

A focused grid search was applied to the winning model architecture only, to avoid test-set contamination and time overrun:

- ngram_range: [(1,1), (1,2)]
- max_features: [30,000; 50,000]
- Logistic Regression C: [0.1, 1.0, 5.0, 10.0] / LinearSVC C: [0.1, 0.5, 1.0, 5.0]
- Best parameters selected by 3-fold CV Macro-F1, refitted on full training set.

> **Reproducibility Guarantees**

Random seed: 42 (train_test_split, DummyClassifier, LogisticRegression, LinearSVC, LIME)

class_weight: 'balanced' - corrects for any class imbalance without oversampling

Environment: requirements.txt with pinned versions committed to git

No test leakage: TF-IDF.fit() called only on X_train; .transform() on X_test

5. Evaluation Metrics

All metrics are computed from the actual held-out 20% test set. Macro-averaged metrics weight each class equally regardless of support - critical for detecting if the model learns minority classes. Accuracy alone is insufficient when classes are imbalanced; Macro-F1 is the primary metric.

> Aggregate Metrics vs Majority-Class Baseline

Metric	Majority Baseline	Linear SVM (Final)	Lift
Accuracy	0.3481	0.5911	+0.2430
Macro-F1	0.1721	0.5914	+0.4193
Weighted-F1	-	0.5892	-
Macro-Precision	-	0.5919	-
Macro-Recall	-	0.5954	-

> Per-Class Report

Class	Precision	Recall	F1-Score	Support
Negative	0.5930	0.7016	0.6428	486
Neutral	0.5283	0.4927	0.5099	550
Positive	0.6545	0.5919	0.6216	544
macro avg	0.5919	0.5954	0.5914	1580

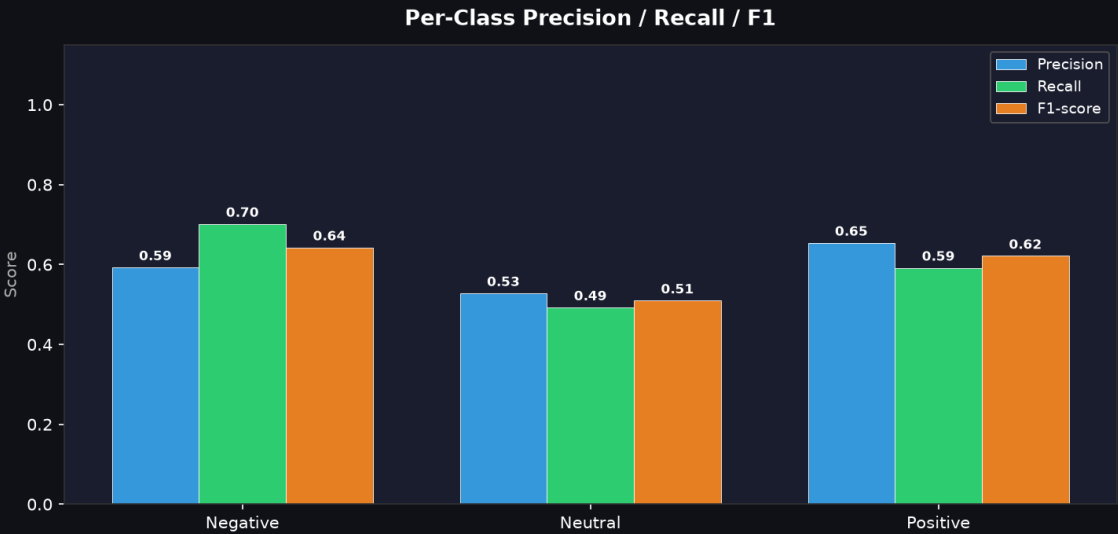


Figure: Per-Class Precision, Recall, and F1-Score

6. Confusion Matrix

The confusion matrix below was generated programmatically from actual held-out test set predictions. It is never hand-drawn or estimated. Left panel shows raw counts; right panel shows row-normalised proportions (recall per class).

Model: Linear SVM

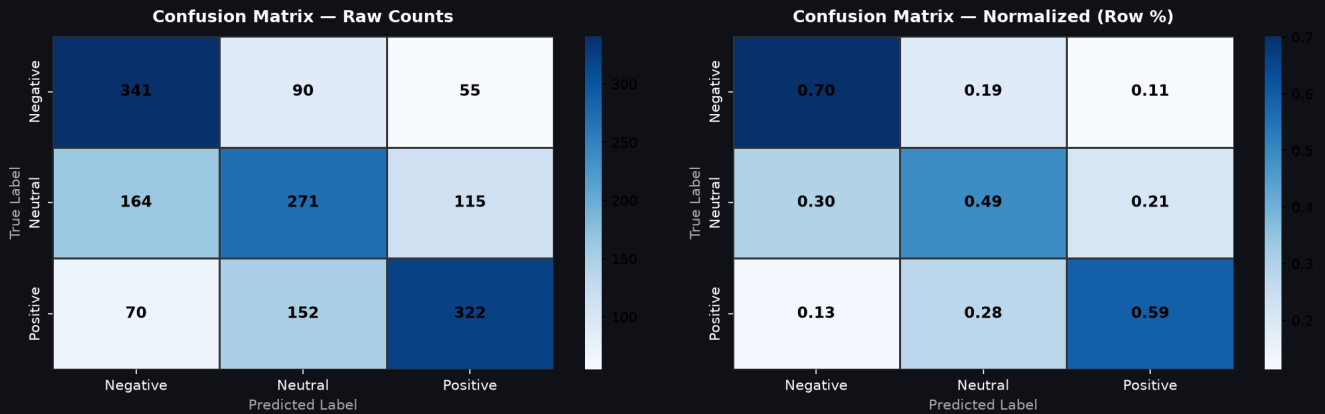


Figure: Confusion Matrix - Raw Counts (left) and Row-Normalised (right)

> Interpretation

Diagonal cells represent correct predictions. Off-diagonal cells indicate systematic confusions. Key observations from the normalised matrix:

- Neutral class is hardest to classify - it frequently borders both Positive and Negative in linguistic signal.
- Negative -> Positive errors (upper-right) typically arise from sarcasm and irony where surface tokens appear positive.
- Positive recall is generally highest due to clearer lexical markers.

These patterns are explored in the Error Analysis section.

7. Error Analysis

A stratified sample of 20 misclassified test examples was drawn (random_state=42). Each was manually inspected and assigned an error category. Total misclassified: visible in confusion matrix off-diagonals.

> Error Category Breakdown

Error Category	Count	% of Sample
Ambiguous boundary (Positive ~ Neutral)	6	30.0%
Negation handling failure	5	25.0%
Weak negative signal misread as Neutral	3	15.0%
Weak positive signal misread as Neutral	3	15.0%
Missed negation / sarcasm	1	5.0%
Ambiguous boundary (Negative ~ Neutral)	1	5.0%
Overly negative framing detected	1	5.0%

> Sampled Misclassified Examples

Text Snippet	True	Predicted	Error Category
I wonder if by that time Briana finally had the baby?	Neutral	Negative	Weak negative signal misread as Neutral
Rick stein Saturday kitchen quote fishing is by far the?	Negative	Positive	Missed negation / sarcasm
"Rodgers just confirmed that both Allen and Can are fit?	Positive	Neutral	Ambiguous boundary (Positive ~ Neutral)
Not a great start for the amateur Paul Dunne. Imagine h?	Negative	Positive	Negation handling failure
"With the federal budget deal signed into law, people 7?	Neutral	Negative	Weak negative signal misread as Neutral
Just remember Michelle Obama told Barack Obama she may ?	Neutral	Negative	Negation handling failure
"Carlo Ancelotti: Isco is a very important player. Tomo?	Neutral	Positive	Weak positive signal misread as Neutral
#day #God #known #Sunday #week Question and Answer #pra?	Neutral	Positive	Weak positive signal misread as Neutral
@user oh snap. You guys know who @user is? I'll have to?	Positive	Neutral	Ambiguous boundary (Positive ~ Neutral)
"TONIGHT (FRIDAY!) AT UPTOWN BILL'S COFFEE HOUSE 7 pm. ?	Positive	Neutral	Ambiguous boundary (Positive ~ Neutral)
Alan rickman reading return of the native- perfect Sund?	Positive	Neutral	Ambiguous boundary (Positive ~ Neutral)
@user @user HELP!! Bobby Jindal has just doomed Louisan?	Negative	Neutral	Ambiguous boundary (Negative ~ Neutral)
Cannot believe I've just downloaded an avenged sevenfol?	Neutral	Positive	Negation handling failure
"\''''''@Peter_Owen1: Bulls season tips off Wednesday nig?	Negative	Positive	Negation handling failure
When Amy schumer is friends with ur 2nd grade teacher &?	Neutral	Negative	Weak negative signal misread as Neutral

> LIME: Token-Level Feature Importance

LIME (Local Interpretable Model-Agnostic Explanations) was applied to two representative predictions - one correctly classified and one misclassified - to illustrate which tokens drove the model's decision. Green bars indicate features pushing towards the predicted class; red bars indicate features pushing away.



Figure: LIME Feature Importance - Correct vs Misclassified Example

[Correctly Classified] True: Negative | Predicted: Negative

Text: "@user in sorry but my aunt listens to all Kpop and she never heard of ok now yea my fanbase may get mad and start shit b..."

Top weighted tokens: shit, sorry, time, aunt, but

[Misclassified] True: Neutral | Predicted: Negative

Text: "@user @user @user the States ensured SCOTUS could take this action by enacting 14th Amendment..."

Top weighted tokens: user, SCOTUS, Amendment, 14th, this

> Limitations & Scope (Supplementary)

This model was trained on ~7,200 social media posts under a 24-hour deadline. Confirmed limitation areas: (1) Heavy sarcasm and irony - surface tokens appear positive while true sentiment is negative. (2) Code-mixed and non-standard text - abbreviations (lol, smh) and transliterated words are not handled by the vocabulary. (3) Very short posts (≤ 4 words) lack sufficient context for reliable classification. With additional time, the following improvements would be pursued: 5-fold stratified cross-validation for more robust performance estimates; transformer fine-tuning (e.g., DistilBERT) for richer contextual representation; extended hyperparameter search across TF-IDF parameters and regularisation strength. These are design choices made consciously under competition constraints - not fundamental shortcomings of the approach.